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Implementation of XGBoost Technique on various health parameters to provide a health score with analysis

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Abstract- ***In today's world wearable sensors play an important role in real time monitoring and early detection of health anomalies .In this project machine learning is implemented to continuously track vital health parameters with the help of which anomalies can be detected. The collected data is processed by ml algorithms (XGBoost) .This is then displayed on a website where users can monitor their health status through a health score metric and different graph visualizations. By combining electronics, machine learning and web based monitoring this system offers a smart and efficient approach to personalized healthcare. Parameters such as SpO2,heart rate, temperature, age and consciousness are taken into consideration to provide a health score analysis.***

Keywords-machine learning, health monitoring device, web based monitoring

I. INTRODUCTION

The advancements in sensor technology have significantly improved real time health monitoring. Wearable and portable health monitoring devices help in continuously monitoring parameters such as heart rate, ECG voltage, blood oxygen levels and body temperature facilitating early diagnosis and timely medical intervention[1].Integration of ml with sensors and microcontrollers will increase the efficiency of health monitoring systems. ML algorithms analyses historical health data to detect anomalies, predict potential health risks, and provide personalized recommendations. Studies have demonstrated that ml models trained on real time sensor data

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can effectively identify patterns indicative of cardiovascular diseases, irregular heart rhythms and other health conditions.

Along with wearable sensors and machine learning the addition of web based implementations have enhanced the efficiency of the system. The use of cloud based storage and web platforms enables secure and scalable data management ensuring that healthcare providers can access patient information from any location.

A. Literature Survey

There have been various studies on sensor-based health monitoring systems, with a focus on their applications in remote healthcare and real-time diagnostic systems. Doe et al. in [1] investigated biomedical sensor technologies for medical care, citing their applications in patient monitoring and disease detection at an early stage. The research focused on the increasing demand for wearable and non-invasive health monitoring devices.

Smith et al. [2] emphasized applying wearable sensors for remote patient monitoring, illustrating how ongoing monitoring of physiological parameters can better enhance patient care. According to their study, integrating multiple readings from sensors increases the trustworthiness of health information and allows real-time diagnosis.

Brown [3] discussed the application of machine learning in healthcare systems and how it can be useful in identifying health anomalies and forecasting medical conditions. The research concluded that ML-based models can enhance diagnostic precision and aid in clinical decision-making based on past health patterns.

Recent studies have also explored web-based implementations of health monitoring. Lee et al. [4] research identified the importance of cloud-based dashboards in telemedicine, enabling healthcare professionals to remotely access patient information. Their results indicated that web integration improves usability and offers real-time feedback into patient health parameters.

Based on these researches, this paper proposes an ml based web-enabled health monitoring system to offer an effective and economical real-time health monitoring solution.
Integrating sensor technology, machine learning and web implementation, the system is able to offer assured and precise health monitoring.
Unlike conventional devices that typically measure one or two parameters this system measures four parameters -pulse, body temperature,SpO2, ECG and it does not require any external shields for connectivity
,all the sensors are directly integrated with ESP 32 ensuring portability and ease of deployment.
So basically in many places linear regression is used as a method to predict the score but here XGBoost is used which is more efficient and accurate.
In most cases only heart rate,spO2 is taken into consideration and health status(positive or negative) is predicted rather than health score, here the patients age, gender and consciousness is also taken into consideration to predict the health score.
Earlier there was no way that a proper analysis along with a health score was displayed. But now with the help of nextjs,rechart.js we can display the health score along with analysis which helps the patient understand any need of immediate action.
Also, there were no options to save data and show doctor what the history of trends of the user's health parameters look like and how they vary. Using a proper backend structure using express.js and MongoDB we can create history trends of health parameters of the user to consult the doctor

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II. METHODOLOGY

A. System Design

The proposed system integrates multiple sensors with ESP 32 microcontroller in order to monitor the vital health parameters in real time. The system architecture is illustrated in **Fig. 1**, which shows the block diagram of the hardware setup. ESP 32 serves as the central processing unit interfacing with the pulse sensor,SpO2 sensor, ECG sensor and temperature sensor. Each sensor is responsible for measuring specific vital parameters as shown in Fig2.Sensor measures the heart rate in beats per minute(bpm),SpO2 sensor measures the SpO2 levels, ECG sensor records the electrical activity of the heart and temperature sensor measures the body temperature. The software implementation involves programming the ESP 32 using Arduino IDE and the collected sensor data is sent to Google sheets.

This data is processed using a machine learning model and a web based dashboard is implemented to provide real-time visualization.

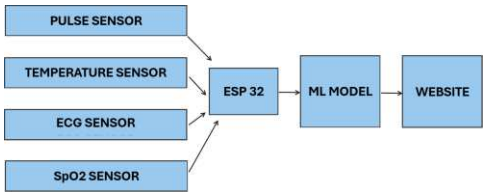


Fig. 1:Block diagram of the system

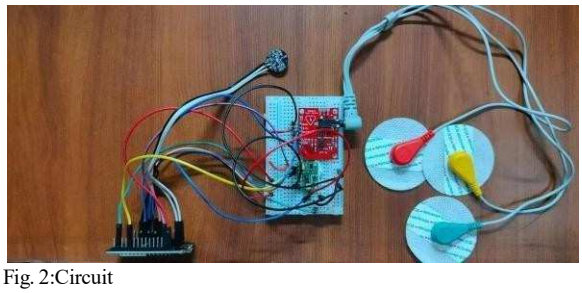


Fig. 2:Circuit

A. Algorithm used

XGBoost is popular algorithm which generates groups of multiple decision trees and continuously learning from mistakes in each round of prediction. The wrong samples from previous samples are given more focus in the succeeding tree. By doing this, the algorithm learns to emphasize the samples that are tougher to classify and gradually improves the accuracy. We're attempting to predict the health score of an individual based on their Heart rate, Body Temperature, Saturation, Consciousness, age and gender. It's objective in the project was to predict the health of a person based on the parameters measured by the setup explained in section A. The dataset used to train the model is shown in **Fig. 4**, containing annotated sample values collected during system development.

Once the model is trained, it can analyze real-time input data and generate health score for the individual, providing valuable insights about potential health risks. The comparison between the actual and predicted health score is shown in **Fig. 3**, indicating the accuracy of the XGBoost model.

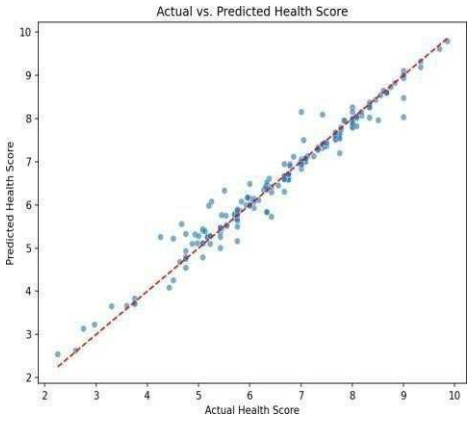


Fig. 3:Plot between actual and predicted health score

HR	TEMP	SPO2	avpu	age	gender	health_score
49.56	36.54	93.18	3	86	0	4.37
60.23	36.39	97.05	1	86	0	8.2
61.28	36.55	101.31	1	86	0	9.2
59.33	36.31	94.47	3	86	0	4.62
63.25	35.54	92.23	2	86	0	5.53
60.85	37.6	97.4	4	86	0	6.2

Fig. 4:Training Dataset

B. Website Implementation

The website integrates the ML model and the values from the hardware model together so that the health score is generated and proper analysis can be done for the health of the given patient. TechStack used for implementing this is Nextjs(Frontend) and for the backend Express.js is being used and all values are being stored in MongoDB database. We fetch the data from the google sheets where our

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hardware system is sending the value then we use the ML model to generate the health score and show analysis of it. Using recharts.js all values can be visualized in real time and user can save all those data in MongoDB and see its graphical visualization . JWT has been used for user authentication. Thus an user can access all data and health score analysis in their respective accounts. The real-time display interface presented to the user upon device connection is shown in **Fig. 5**.

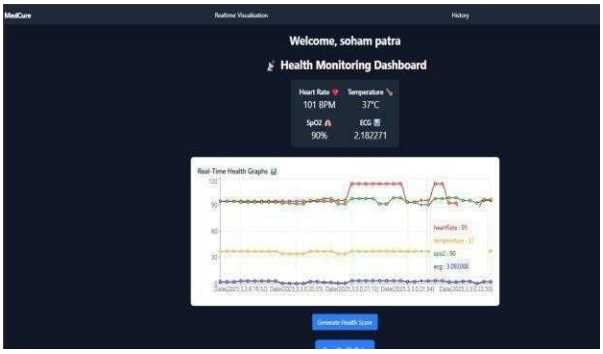


Fig. 5:Real Time data display in the beginning when hardware device is connected

III. RESULT

The system was tested under various conditions . The ECG voltage readings remained within the expected physiological range of 0.5 volt to 2.5 vol. Heart rate values fluctuated between 60 to 100 bpm.SpO2 levels were monitored effectively. Temperature readings were stable within the normal body temperature range. Data transmission to Google sheets was seamless. The machine learning model successfully identified patterns in the recorded data and the web based interface provided a platform for monitoring the health metrics remotely. The real-time data sent from the ESP32 device to Google Sheets is tabulated in **Table 1**, containing heart rate, SpO_2, ECG, and temperature values across timestamps.

Timestamp	BPM	SpO_2	ECG	Temperature (°C)
02/04/2025 23:06:02	89	96	1.106447	34
02/04/2025 23:06:13	94	98	1.083077	34
02/04/2025 23:06:23	88	94	1.093553	34
02/04/2025 23:06:32	83	93	1.048425	35
02/04/2025 23:06:42	76	90	1.033919	35
02/04/2025 23:06:52	82	97	1.031506	35
02/04/2025 23:07:01	91	96	1.026667	35
02/04/2025 23:07:11	94	95	1.021049	36
02/04/2025 23:07:21	87	98	1.012696	36
02/04/2025 23:07:31	90	94	1.009821	36
02/04/2025 23:07:41	88	97	0.998261	36
02/04/2025 23:07:50	91	98	0.980733	35
02/04/2025 23:08:01	92	96	0.957363	37
02/04/2025 23:08:12	92	96	0.957363	37
02/04/2025 23:08:22	94	97	0.946886	34

Table 1:data sent to google sheets

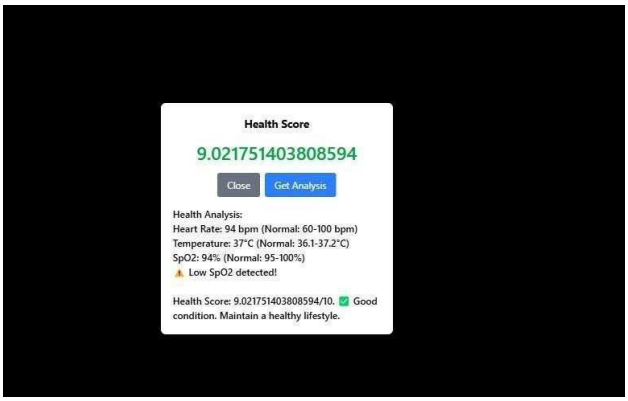


Fig. 6:HealthScore Generated with analysis

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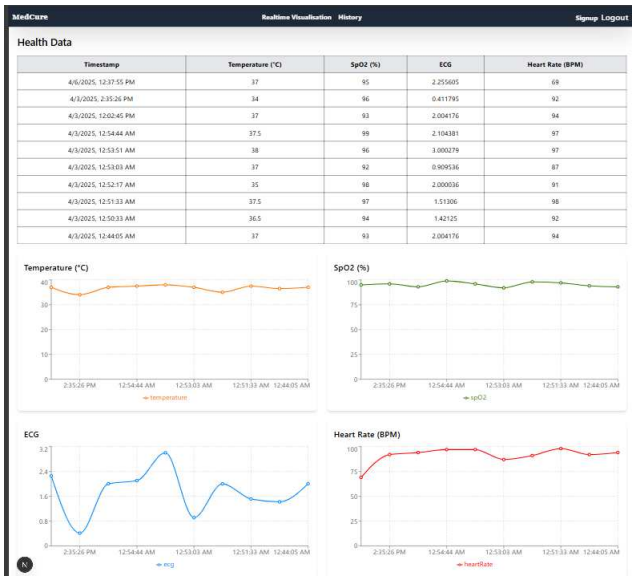


Fig. 7:Visualization of Saved Health Data

IV. CONCLUSION

The health device consisting of the heart rate,spO2,ECG,temperature sensor ,which is built in a cost-effective way yet it is very accurate in measuring the values. Given our ML model which uses XG Boost technique which is a proficient method to take all attributes into account and produce a very accurate health score which helps patients along with doctors to keep monitoring the health on a regular basis. This helps in raising alerts and reaching out to doctors whenever the need arises at an early stage and thus curing severe diseases which could have happened in the absence of an early treatment. Hence this will improve the healthcare facilities of the nation. The system generated health score with its accompanying analysis is shown in **Fig 6**.The visualization of saved health data over time for trend analysis is depicted in **Fig 7**, enabling a doctor or patient to track variations.

V. COMPARISON WITH EARLIER WORKS

A comparative evaluation of prior systems versus the proposed solution is shown in **Table 2**, highlighting improvements across machine learning, data storage, visualization, and clinical readiness

Criteria	**Earlier Works**	**This Work**
Sensor Integration	Typically used basic sensors (heart rate, SpO_2) with limited data points. Often required additional hardware for communication.	Integrates four sensors (Heart rate, SpO_2, ECG, Temperature) directly into ESP32, eliminating external shields. This makes it compact, portable, and cost-efficient.
Machine Learning Technique	Mostly relied on Linear Regression, SVM, or simple rule-based classification for predicting binary health status (healthy/unhealthy).	Uses XGBoost, a powerful ensemble model that iteratively improves its predictions. This allows for more accurate, nuanced health score prediction rather than binary classification.
Health Score Generation	Most systems only classify health into positive/negative or high/low risk. Age, gender, or consciousness were not usually considered.	Predicts a quantitative health score based on heart rate, SpO_2, temperature, consciousness, age, and gender, offering personalized, multifactorial insight.
Data Storage & History Tracking	Rarely supported history tracking or used simple local storage.	Uses MongoDB to store long-term data for each user. Enables tracking trends over time and helps with doctor consultations.
Platform Architecture	Either limited to embedded devices or disconnected apps; not seamlessly integrated.	Complete full-stack implementation using Express.js (backend) and Next.js (frontend) for a holistic, web-based, real-time system.
Data Flow & Pipeline	Siloed or semi-automated pipelines with delays in ML integration.	End-to-end pipeline: ESP32 → Google Sheets → ML Model → Web Dashboard, forming a real-time, closed feedback loop.

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